Arjuna JEE 2027

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

TRIGONOMETRIC FUNCTIONS

DIBY (DO IT BY YOURSELF)

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DIBY-01

If $\sin A \tan A = \cos^2 A$, then $\cos^3 A + \cos^2 A$ is equal to

(A) 1 (B) 2 (C) 4 (D) None of these

[Ans. : (A)]

DIBY-02

If $\text{cosec}A + \cot A = \frac{11}{2}$, then $\tan A$ is

(A) $\frac{21}{22}$ (B) $\frac{15}{16}$ (C) $\frac{44}{117}$ (D) $\frac{117}{43}$

[Ans. : (C)]

DIBY-03

If $\tan\theta = \frac{-4}{3}$, then $\sin\theta$ is

(A) $\frac{-4}{5}$ but not $\frac{4}{5}$ (B) $\frac{-4}{5}$ or $\frac{4}{5}$ (C) $\frac{4}{5}$ but not $-\frac{4}{5}$ (D) None of these

[Ans. : (B)]

DIBY-04

If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, then prove that $m^2 - n^2 = 4\sin\theta\tan\theta$

DIBY-05

Prove that $\frac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \frac{1 + \sin A}{\cos A}$

DIBY-06

Prove that: $\sin^8 A - \cos^8 A = (\sin^2 A - \cos^2 A)(1 - 2\sin^2 A\cos^2 A)$

DIBY-07

$(1 + \tan\alpha\tan\beta)^2 + (\tan\alpha - \tan\beta)^2 =$

(A) $\tan^2\alpha\tan^2\beta$

(B) $\sec^2\alpha\sec^2\beta$

(C) $\tan^2\alpha\cot^2\beta$

(D) $\sec^2\alpha\cos^2\beta$

[Ans. : (B)]

DIBY-08

Prove that : $\frac{\sin A}{1+\cos A}+\frac{1+\cos A}{\sin A}=2\operatorname{cosec} A$

DIBY-09

Prove that : $\sqrt{\frac{1-\sin A}{1+\sin A}}=\sec A-\tan A$

DIBY-10

Prove that : $\frac{\operatorname{cosec} A}{\operatorname{cosec} A-1}+\frac{\operatorname{cosec} A}{\operatorname{cosec} A+1}=2\sec^2 A$

DIBY-11

Prove that : $\frac{\operatorname{cosec} A}{\cot A+\tan A}=\cos A$

DIBY-12

Find value of following

(i) $\sin^2\frac{\pi}{6}+\cos^2\frac{\pi}{3}-\tan^2\frac{\pi}{4}=$

(ii) $2\sin^2\frac{\pi}{6}+\operatorname{cosec}^2\frac{7\pi}{6}\cos^2\frac{\pi}{3}=$

(iii) $\cot^2\frac{\pi}{6}+\operatorname{cosec}\frac{5\pi}{6}+3\tan^2\frac{\pi}{6}=$

(iv) $2\sin^2\frac{3\pi}{4}+2\cos^2\frac{\pi}{4}+2\sec^2\frac{\pi}{3}=$

[Ans. : (i) $\frac{-1}{2}$, (ii) $\frac{3}{2}$, (iii) 6, (iv) 10]

DIBY-13

$\frac{\tan(90°-\theta)\sec(180°-\theta)\sin(-\theta)}{\sin(180°+\theta)\cos(360°+\theta)\operatorname{cosec}(90°-\theta)}$ is equals

(A) $\operatorname{cosec}\theta$ (B) 1 (C) –1 (D) $-\operatorname{cosec}\theta$

[Ans. : (D)]

DIBY-14

$\sec(270^\circ - A)\sec(90^\circ - A) - \tan(270^\circ - A)\tan(90^\circ + A)$

[Ans. : –1]

DIBY-15

$\cot A + \tan(180^\circ + A) + \tan(90^\circ + A) + \tan(360^\circ - A)$

[Ans. : 0]

DIBY-16

$$\frac{\tan(180^\circ-\theta)\cos(180^\circ-\theta)\tan(90^\circ-\theta)}{\sin(90^\circ+\theta)\cot(90^\circ-\theta)\tan(90^\circ+\theta)}$$

[Ans. : –1]

DIBY-17

If $\tan A = 2$ and $A \in \left(\pi, \frac{3\pi}{2}\right)$ then the expression $\frac{\cos A}{\sin^3 A + \cos^3 A}$ is equal to

[Ans. : 5/9]

DIBY-18

$\sin^2\frac{\pi}{18}+\sin^2\frac{\pi}{9}+\sin^2\frac{7\pi}{18}+\sin^2\frac{4\pi}{9}$ equal

(A) 0 (B) 1 (C) –1 (D) 2

[Ans. : (D)]

DIBY-19

Prove that

1. $\tan 720^\circ-\cos 270^\circ-\sin 150^\circ\cos 120^\circ=\frac{1}{4}$
2. $\sin 780^\circ\sin 480^\circ+\cos 120^\circ\sin 150^\circ=\frac{1}{2}$

DIBY-20

Prove that $\sec\left(\frac{3\pi}{2}-\theta\right)\sec\left(\theta-\frac{5\pi}{2}\right)+\tan\left(\frac{5\pi}{2}+\theta\right)\tan\left(\theta-\frac{3\pi}{2}\right)=-1$.

DIBY-21

If $\sin x = -\frac{3}{5}$ where $\pi < x < \frac{3\pi}{2}$, then $80(\tan^2 x - \cos x)$ is equal to

(A) 109 (B) 108 (C) 19 (D) 18

[Ans. : (A)]

DIBY-22

1. $\sin(405^\circ)$ [Ans. : $\frac{1}{\sqrt{2}}$]

2. $\sec(-1470^\circ)$ [Ans. : $\frac{2}{\sqrt{3}}$]

3. $\tan(-300^\circ)$ [Ans. : $\sqrt{3}$]

4. $\cot(585^\circ)$ [Ans. : 1]

5. $\operatorname{cosec}(-750^\circ)$ [Ans. : –2]

6. $\cos(-2220^\circ)$ [Ans. : 1/2]

DIBY-23

Prove that

1. $\frac{\sin 7\theta-\sin 5\theta}{\cos 7\theta+\cos 5\theta}=\tan\theta$

2. $\frac{\cos 6\theta-\cos 4\theta}{\sin 6\theta+\sin 4\theta}=-\tan\theta$

3. $\frac{\sin A+\sin 3A}{\cos A+\cos 3A}=\tan 2A$

4. $\frac{\sin 7A-\sin A}{\sin 8A-\sin 2A}=\cos 4A\sec 5A$

5. $\frac{\cos 2B+\cos 2A}{\cos 2B-\cos 2A}=\cot(A+B)\cot(A-B)$

6. $\frac{\sin 2A+\sin 2B}{\sin 2A-\sin 2B}=\frac{\tan(A+B)}{\tan(A-B)}$

7. $\frac{\sin A+\sin 2A}{\cos A-\cos 2A}=\cot\frac{A}{2}$

8. $\frac{\sin 5A-\sin 3A}{\cos 3A+\cos 5A}=\tan A$

9. $\frac{\cos 2B-\cos 2A}{\sin 2B+\sin 2A}=\tan(A-B)$

10. $\cos(A+B)+\sin(A-B)=2\sin(45^\circ+A)\cos(45^\circ+B)$

11. $\frac{\cos 3A-\cos A}{\sin 3A-\sin A}+\frac{\cos 2A-\cos 4A}{\sin 4A-\sin 2A}=\frac{\sin A}{\cos 2A\cos 3A}$

DIBY-24

Find the value of expression $\frac{\sin 8\theta\cos\theta - \sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta - \sin 3\theta\sin 4\theta}$, given $\theta = 7.5°$.

(A) $2 + \sqrt{3}$ (B) $2 - \sqrt{3}$ (C) 0 (D) 1

[Ans. : (B)]

DIBY-25

If $\alpha = \frac{\pi}{19}$, find the value of $\frac{\sin 23\alpha - \sin 3\alpha}{\sin 16\alpha + \sin 4\alpha}$.

(A) 1 (B) 0 (C) 2 (D) –1

[Ans. : (D)]

DIBY-26

The expression $\frac{(1+\tan 37°)(1+\tan 8°)(1+\tan 15°)(1+\tan 30°)}{(1+\tan 2°)(1+\tan 43°)}$ reduces to

(A) 2 tan 45° (B) 4 (C) 8 (D) 6 cot 45°

[Ans. : (A)]

DIBY-27

For $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, let $3\sin(\alpha+\beta) = 2\sin(\alpha-\beta)$ and a real number k be such that $\tan\alpha = k\tan\beta$. Then the value of k is equal to

(A) 5 (B) $-2/3$ (C) -5 (D) $2/3$

[Ans. : (C)]

DIBY-28

Let $\frac{\tan 50^\circ - \tan 40^\circ}{\tan 10^\circ} = k$, find $k \in N$.

[Ans. : 2]

DIBY-29

If $x = \frac{\pi}{48}$, find the value of $\frac{\sin 7x + \sin 5x + \sin 9x + \sin 3x}{\cos 7x + \cos 5x + \cos 9x + \cos 3x}$.

[Ans. : $\sqrt{2} - 1$]

DIBY-30

If $3\sin\alpha = 5\sin\beta \neq 0$, then the value of $\frac{\tan\left(\frac{\alpha+\beta}{2}\right)}{\tan\left(\frac{\alpha-\beta}{2}\right)}$ is:

[Ans. : 4]

DIBY-31

The value of $\cos 24^\circ - \cos 12^\circ + \cos 48^\circ - \cos 84^\circ =$

[Ans. : 1/2]

DIBY-32

If $\cos\alpha + \cos\beta = \frac{1}{2}$ and $\sin\alpha + \sin\beta = \frac{1}{3}$

Column - I	Column - II
(A) $\cos\left(\frac{\alpha+\beta}{2}\right)$	(P) $\pm\frac{\sqrt{13}}{12}$
(B) $\cos\left(\frac{\alpha-\beta}{2}\right)$	(Q) $\frac{2}{3}$
(C) $\tan\left(\frac{\alpha+\beta}{2}\right)$	(R) $\pm\frac{3}{\sqrt{13}}$
(D) $\tan\left(\frac{\alpha-\beta}{2}\right)$	(S) $\pm\sqrt{\frac{131}{13}}$

[Ans. : A-R,B-P,C-Q,D-S]

DIBY-33

Prove that $\frac{\sin\theta+\sin 2\theta}{1+\cos\theta+\cos 2\theta}=\tan\theta$

DIBY-34

Prove that $\frac{1-\cos A+\cos B-\cos(A+B)}{1+\cos A-\cos B-\cos(A+B)}=\tan\frac{A}{2}\cot\frac{B}{2}$

DIBY-35

Value of : $\prod_{k=1}^{17}\sin\frac{k\pi}{18}=\frac{a}{2^{16}}$ then $a=$ _____.

[Ans. : 9]

DIBY-36

If $\frac{5\pi}{2}<x<3\pi$, then the value of the expression $\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}}$ is

(A) $-\cot\frac{x}{2}$ (B) $\cot\frac{x}{2}$ (C) $\tan\frac{x}{2}$ (D) $-\tan\frac{x}{2}$

[Ans. : (D)]

DIBY-37

If $A = 340^\circ$ then $2\sin\frac{A}{2}$ is identical to

(A) $\sqrt{1+\sin A}+\sqrt{1-\sin A}$ (B) $-\sqrt{1+\sin A}-\sqrt{1-\sin A}$

(C) $\sqrt{1+\sin A}-\sqrt{1-\sin A}$ (D) $-\sqrt{1+\sin A}+\sqrt{1-\sin A}$

[Ans. : (D)]

DIBY-38

Evaluate: $\cos^6\frac{\pi}{16}+\cos^6\frac{3\pi}{16}+\cos^6\frac{5\pi}{16}+\cos^6\frac{7\pi}{16}$

(A) 5/8 (B) 5/4 (C) 7/16 (D) 5/32

[Ans. : (B)]

DIBY-39

If $\frac{x}{\cos\theta}=\frac{y}{\cos\left(\theta+\frac{2\pi}{3}\right)}=\frac{z}{\cos\left(\theta-\frac{2\pi}{3}\right)}$, then $x+y+z$ is equal to:

(A) –1 (B) 1 (C) 0 (D) 2

[Ans. : (C)]

DIBY-40

Find the exact value of $\tan^2\frac{\pi}{16}+\tan^2\frac{3\pi}{16}+\tan^2\frac{5\pi}{16}+\tan^2\frac{7\pi}{16}$

[Ans. : 28]

DIBY-41

Prove that $\cos36^\circ\cos72^\circ\cos108^\circ\cos144^\circ = 1/16$.

DIBY-42

If $A+B+C=\pi$, prove that $\sin^2\frac{A}{2}+\sin^2\frac{B}{2}-\sin^2\frac{C}{2}=1-2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$.

DIBY-43

If $A+B+C=180^\circ$, prove that $\cos^2A+\cos^2B+\cos^2C=1-2\cos A\cos B\cos C$.

DIBY-44

In triangle ABC, prove that

$$\sin(B+C-A)+\sin(C+A-B)+\sin(A+B-C)=4\sin A\,\sin B\,\sin C.$$

DIBY-45

If $A+B+C=\pi/2$, show that

(a) $\sin^2 A+\sin^2 B+\sin^2 C=1-2\sin A\,\sin B\,\sin C$

(b) $\cos^2 A+\cos^2 B+\cos^2 C=2+2\sin A\,\sin B\,\sin C$

DIBY-46

If $\frac{\sin^4 x}{2}+\frac{\cos^4 x}{3}=\frac{1}{5}$**, then** **[IIT-JEE-2009]**

(A) $\tan^2 x=\frac{2}{3}$

(B) $\frac{\sin^8 x}{8}+\frac{\cos^8 x}{27}=\frac{1}{125}$

(C) $\tan^2 x=\frac{1}{3}$

(D) $\frac{\sin^8 x}{8}+\frac{\cos^8 x}{27}=\frac{2}{125}$

[Ans. : A, B]

DIBY-47

For $\triangle ABC$, $\sin\frac{A}{2}+\sin\frac{B}{2}+\sin\frac{C}{2}-1=\lambda\sin\frac{\pi-A}{4}\sin\frac{\pi-B}{4}\sin\frac{\pi-C}{4}$. Then $\lambda=$

[Ans. : 4]

DIBY-48

The value of $\sin^3 10^\circ + \sin^3 50^\circ - \sin^3 70^\circ$ is equal to

(A) $-\frac{3}{2}$ (B) $\frac{3}{4}$ (C) $-\frac{3}{4}$ (D) $-\frac{3}{8}$

[Ans. : (D)]

DIBY-49

If $\frac{\sin^4 x}{2}+\frac{\cos^4 x}{3}=\frac{1}{5}$ then the value of $\cot^2 x$ is equal to _____.

[Ans. : 3/2]

DIBY-50

The value of $\frac{\tan(\ln 6)-\tan(\ln 2)-\tan(\ln 3)}{\tan(\ln 6)\cdot\tan(\ln 2)\cdot\tan(\ln 3)}$ is equal to _____.

[Ans. : 1]

DIBY-54

Find the summation of the following series

(i) $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}$

(ii) $\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{5\pi}{7}+\cos\frac{6\pi}{7}$

[Ans. : (i) -1/2 (ii) 0]

DIBY-55

Prove that $\sin\theta+\sin 3\theta+\sin 5\theta+\cdots+\sin(2n-1)\theta=\frac{\sin^2 n\theta}{\sin\theta}$

DIBY-56

Sum the series : $\sqrt{1+\cos\alpha}+\sqrt{1+\cos 2\alpha}+\sqrt{1+\cos 3\alpha}+\cdots$ **to** n **terms, where** $0<\alpha<\pi$

$$\left[\text{Ans. } \sqrt{2}\,\frac{\sin\frac{n\alpha}{4}}{\sin\frac{\alpha}{4}}\cos\left[(n+1)\frac{\alpha}{4}\right]\right]$$

DIBY-57

Find the value of $(\cos^4 1^\circ + \cos^4 2^\circ + \cos^4 3^\circ + \cdots + \cos^4 179^\circ) - (\sin^4 1^\circ + \sin^4 2^\circ + \sin^4 3^\circ + \cdots + \sin^4 179^\circ)$.

[Ans. : -1]

DIBY-58

If $\frac{\sin\theta}{\cos(3\theta)} + \frac{\sin(3\theta)}{\cos(9\theta)} + \frac{\sin(9\theta)}{\cos(27\theta)} + \frac{\sin(27\theta)}{\cos(81\theta)} = \frac{\sin(k\theta)}{2\cos\theta\cos 81\theta}$, then k equals to

[Ans. : 80]

DIBY-59

$\sin\theta \cdot \sec 3\theta + \sin 3\theta \cdot \sec 3^2\theta + \sin 3^2\theta \cdot \sec 3^3\theta + \cdots$ upto n terms $=$

(A) $\frac{1}{2}[\tan 3^n\theta - \tan\theta]$

(B) $\frac{1}{2}[\tan 3^n\theta + \tan\theta]$

(C) $[\tan 3^n\theta + \tan\theta]$

(D) $[\tan 3^n\theta - \tan\theta]$

[Ans. : (A)]

DIBY-60

If m and n are positive integers satisfying $1+\cos 2\theta+\cos 4\theta+\cos 6\theta+\cos 8\theta+\cos 10\theta=\frac{\cos m\theta \cdot \sin n\theta}{\sin \theta}$, then $m+n$ is equal to

[Ans. : 11]

DIBY-61

Find the sum of series $S=\sin\theta-\sin 2\theta+\sin 3\theta-\sin 4\theta+\cdots+n$ terms.

[Ans. : .....]

DIBY-62

The value of

$P=(\sqrt{3}+\tan 1^\circ)(\sqrt{3}+\tan 2^\circ)(\sqrt{3}+\tan 3^\circ)\ldots(\sqrt{3}+\tan 28^\circ)(\sqrt{3}+\tan 29^\circ)$ is

(A) 2^{28} (B) $2^{15}+2-\sqrt{3}$ (C) 2^{29} (D) $2^{14}+2-\sqrt{3}$

[Ans. : (C)]

It is never about end result

It's always about Journey

THANK You

#futureIITians